A High-Performance Document Image Binarization Pipeline with Parallel Image Processing

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ABSTRACT

This is a placeholder abstract for the paper.

KEYWORDS

document image binarization, image processing, OCR preprocessing, adaptive thresholding

1 INTRODUCTION

This is a placeholder for the introduction section.

The article template's documentation, available at https://www.acm.org/publications/proceedings-template, has a complete explanation of these commands and tips for their effective use.

Note that authors' addresses are mandatory for journal articles.

2 RIGHTS INFORMATION

Placeholder for rights information.

3 CCS CONCEPTS AND USER-DEFINED KEYWORDS

Placeholder for CCS concepts.

4 SECTIONING COMMANDS

Placeholder for sectioning commands.

4.1 Subsection

This is a placeholder subsection.

4.1.1 Subsubsection. This is a placeholder subsubsection.

Paragraph. This is a placeholder paragraph.
Subparagraph This is a placeholder subparagraph.

5 TABLES

6 MATH EQUATIONS

6.1 Inline (In-text) Equations

Placeholder for inline equations section.

6.2 Display Equations

Placeholder for display equations section.

$$x = y + z \tag{1}$$

Table 1: Table placeholder

Column 1	Column 2	Column 3
Data 1	Data 2	Data 3

Figure 1: Figure caption placeholder.

7 FIGURES

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

7.1 The "Teaser Figure"

A "teaser figure" is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
  \Description{figure description}
\end{teaserfigure}
```

8 CITATIONS AND BIBLIOGRAPHIES

The use of BibTEX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

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The command `\addbibresource{bibfile}` declares the BibTEX file to use in the **preamble** (before the command "`\begin{document}`") of your LATEX source where "`bibfile`" is the name, *with* the "`.bib`" suffix. Notice that `\addbibresource` takes only one argument: to declare multiple files, use multiple instances of the command.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the "author year" style; for these exceptions, please pass the option `style=acmauthoryear` to the `biblatex` package loaded in the **preamble** (before the command "`\begin{document}`") of your LATEX source.

Some example citations [Abril and Plant(2007)].

9 ACKNOWLEDGMENTS

Placeholder for acknowledgments.

10 APPENDICES

Placeholder for appendices.

11 MULTI-LANGUAGE PAPERS

Placeholder for multi-language papers section.

12 SIGCHI EXTENDED ABSTRACTS

Placeholder for SIGCHI Extended Abstracts section.

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REFERENCES

[Abril and Plant(2007)] Patricia S. Abril and Robert Plant. 2007. The patent holder's dilemma: Buy, sell, or troll? *Commun. ACM* 50, 1 (Jan. 2007), 36–44. https://doi.org/10.1145/1188913.1188915

A RESEARCH METHODS

A.1 Part One

Placeholder for Part One.

A.2 Part Two

Placeholder for Part Two.

B ONLINE RESOURCES

Placeholder for Online Resources.